Deterministic Timing Report

Software simulation evidence for the Embedded Real-Time Motion Controller. This report does not claim real hardware EtherCAT/CANopen validation or certified safety.

Timing Summary

Metric	Value
Cycle count	2500
Min cycle ns	990000
Max cycle ns	3000000
Mean cycle ns	1001296
Min jitter ns	-10000
Max jitter ns	2000000
Mean abs jitter ns	6536
Missed cycles	2
Deadline overruns	2

Motion Performance

Metric	Value
Target position	1.0
Final position	1.0
Overshoot	0.0
Settling time s	0.8
Steady-state error	0.0
Max following error	0.0
Two-axis RMS path error	0.0
Two-axis max path error	0.0

Bus Diagnostics

Counter	Value
TX packets	2500
RX packets	2494
Packet loss count	6
Dropout count	0
Invalid response count	0
Heartbeat timeout count	0
Cable disconnect events	0

Fault Recovery Performance

Metric	Value
Fault observed	True
Operation re-enabled	True
Post-recovery settled	True
Detection latency cycles	0
Recovery cycles	7
Baseline settling s	0.791
Post-recovery settling s	0.441
Settling-time ratio	0.557522
Post-recovery final error	0.0

Jitter Distribution

Jitter bucket	Cycles
-10us	120
-9us	119
-8us	119
-7us	119
-6us	119
-5us	118
-4us	119
-3us	119
-2us	119
-1us	119
+0us	118
+1us	119
+2us	119
+3us	119
+4us	119
+5us	119
+6us	119
+7us	119
+8us	119
+9us	119
+10us	119
+1245us	1
+2000us	1

Scenario Pass/Fail

Scenario	Result	Recovery cycles
normal startup	PASS	0
drive enable sequence	PASS	0
absolute move	PASS	0
relative move	PASS	0
velocity move	PASS	0
quick stop during motion	PASS	73
E-stop safe stop during motion	PASS	75
fault reset and recovery	PASS	5
invalid state transition rejected	PASS	1
missed-cycle injection	PASS	0
CPU load jitter stress	PASS	0
heartbeat timeout	PASS	5
bus cable disconnect simulation	PASS	1
packet loss simulation	PASS	0
encoder feedback loss	PASS	4
following error	PASS	1
command abuse during motion	PASS	1
sensor noise injection	PASS	0
warm restart	PASS	5
cold restart	PASS	5
persistent fault log replay	PASS	0